

Foresight - Lessons from Social Epistemology

Listen. There is something here that is not merely a methodological detail. Once you put people together to produce knowledge about futures, you have made an epistemic arrangement, whether this was what you thought you were doing or not. Who speaks, who is listened to, what is fed back, what counts as a reason, who changes estimate after seeing others - all this starts to matter. Group does not merely collect knowledge that was already there. It does something to it.

Social epistemology has discussed such problems for long time. Foresight has mostly met them in more practical language: bias, groupthink, participation, expert judgement, consensus, diversity, and so on. Fine. But behind these are older and more annoying questions about knowledge. And some things that look obviously good in a workshop become less obviously good once these questions are taken seriously. Six of them are enough here.

1. Convergence is not the same as convergence on the truth. First round, estimates run from 0.05 to 0.85. Third round, everybody sits near 0.35. The panel has converged. Good. But why exactly is 0.35 epistemically better than the disagreement that was there before?

The *Condorcet Jury Theorem* gives one reason for optimism about groups. If each voter is more likely than not to be right, and if votes are independent and sincere, then probability that majority is right increases with size of group (Condorcet, 1785). But independence is doing work there. Feedback rounds make people less independent from each other, which is partly the whole point of feedback. People hear arguments, see distributions, perhaps notice they are outlier, perhaps become less sure. Some of this is learning. Some may be imitation. Usually one does not know where one ends and other begins.

Final number does not tell this. It tells where people ended. It does not tell why they ended there. So convergence is evidence of convergence, certainly. It is not by itself evidence that group became more correct. *The simple move from agreement to epistemic progress is not available.*

2. How disagreement is handled is already a philosophical decision. Suppose protocol says that those with extreme estimates must explain themselves. This sounds normal enough. But already there is something odd: why exactly must the outlier explain, while those sitting near median can remain quiet?

There is a large debate in epistemology of disagreement about this. The *Equal Weight View* says, roughly, that when an epistemic peer disagrees with you, the disagreement itself gives reason to lower confidence (Christensen, 2007). If you would have regarded other person's judgement as good evidence when she agreed with you, you cannot make it worthless only because she disagrees. Then again, Kelly (2010) argues that first-order evidence matters also. Sometimes person who does not move is rational one. Perhaps she simply has better reasons than others.

Now your protocol has to do something, even if nobody involved has read any of this. Outlier explains or does not explain. Group disagreement counts as evidence or it does not count so much. The method cannot stay outside the philosophical issue. It has already taken some position by how the rounds are built. This is one of those places where foresight can think philosophy comes later, but actually *it came earlier and was written into the procedure*.

3. Individually consistent people can still produce a strange group judgement. The classic case is a court with three judges. Liability requires two things: contract forbade an action and defendant performed it (Kornhauser & Sager, 1986). Two judges think contract forbade it. Two think action was performed. So both premises have majority support. Yet only one judge accepts both premises, meaning two judges would reject liability.

Nobody has to reason wrongly. The trouble comes when individual judgements are combined. List and Pettit (2002) show how the problem generalises, under certain conditions, to sets of logically connected propositions.

Foresight does this kind of thing all the time, although language is different. Technology matures, regulation supports it, costs decrease, public acceptance develops, pathway is plausible. These statements are connected, whether questionnaire treats them as separate rows or not. Then we aggregate each answer and call resulting set "group judgement". Maybe it is coherent. Maybe nobody in room actually holds it. Maybe the premises accepted by group do not support conclusion accepted by same group. Putting questions separately in questionnaire does not make them logically independent. *Reasonable individuals do not automatically give you reasonable group*.

4. More communication is not automatically better. This one is uncomfortable because more communication sounds so obviously good. If knowledge is distributed, surely we should make people share more of it, more quickly, through more rounds.

Zollman (2007) gives reasons to be careful. In tightly connected epistemic networks, early evidence can spread quickly. If that evidence happens to support wrong hypothesis, everybody may move to same place before better evidence has time to develop. Communication has worked perfectly in one sense. Information moved fast. This can be exactly why group performs badly.

In a slower or more sparse network some people remain on alternative hypothesis simply because they did not receive everything immediately. That looks inefficient. But if alternative hypothesis turns out to be better one, those people are now useful. Rosenstock et al. (2017) show that effect depends on parameters of model, so one should not make stupid conclusion that less communication is always better. It is not that. But *more communication = epistemically better* is gone. One has to ask what moves, how fast, between whom, and what happens to alternatives while it moves. Amount of communication alone tells rather little.

5. A missing voice can mean missing knowledge, and the worst part is that you may not see the hole. Fricker (2007) calls it *testimonial injustice* when a person is given less credibility than deserved because of prejudice connected to identity. Usually this is discussed as something done to person. For foresight there is also another problem: project may lose something it needed to know.

Suppose an elder knows how local water system has behaved over fifty years. No formal credentials, maybe no numbers, perhaps speaks in way that does not fit vocabulary of project. The knowledge is not taken seriously, and the person is treated as someone whose local knowledge does not really count. Still, person may know where system fails first, what happened in exceptional winter, which local repair changed something, what official record never really captured. If this contribution is discounted, the project does not merely behave unfairly. It may lose access to relevant structure of system.

And this loss is difficult because it does not necessarily leave trace. If a claim is heard and rejected, at least something happened and perhaps can be inspected later. If person was not invited, or stopped speaking because it became obvious what kind of knowledge counts, there is nothing in material saying *important thing missing here*. One cannot miss what was never said.

This is why participation has epistemological side and not only political one. Having many participants is not enough either. The question is what knowledge has possibility to enter and whether it can actually affect anything.

6. Objectivity is not what happens when disagreement disappears. Longino (1990, 2002) argues that assumptions through which data become evidence cannot be checked properly by individual alone. Objectivity depends on critical interaction in community. She gives four norms: recognised forums for criticism, uptake of criticism, public standards for assessing claims, and *tempered equality of intellectual authority*.

Apply these to foresight and some ordinary questions start looking different. Does reasoning circulate or only estimates? If somebody gives a good objection, can it change what group does? Do participants know what counts as good argument? Does somebody's view dominate because person knows more, or because everybody knows the institutional position of that person? These are not exotic questions once put like this.

And notice what follows. A group can agree and still be poor in this sense. Another group may disagree after rather good critical process. So consensus does not solve objectivity problem. *Consensus and objectivity are different things*. Again, something that looked almost self-evident becomes less easy once philosophy has touched it.

Six lessons are obviously not whole social epistemology. There is much more on disagreement, testimony, expertise, aggregation, ignorance, epistemic injustice, networks and collective knowledge. But six is enough to make the point difficult to avoid.

The decisions are made somewhere. Who is invited, what gets circulated, how fast it circulates, whether outlier has to defend herself, whether connected claims are aggregated separately, whose knowledge counts, what criticism can change. These look practical decisions, administrative ones even, and usually they are made as such.

But look again. You decided how disagreement matters. You decided how individual judgements become group judgement. You decided how fast information can pull people toward same view. You decided which knowledge gets chance to enter. You decided what criticism is allowed to do. Maybe all good decisions. Maybe not. That is not the point.

The point is that they were epistemological decisions already. You just called them methodology.

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